NumPy

What is NumPy?

NumPy (short form of **Numerical Python**) is an open-source Python library used for numerical computing. It provides tools for working efficiently with arrays, matrices, and mathematical functions.

NumPy introduces a new data structure called the **ndarray** (n-dimensional array), which is much faster and more memory-efficient than Python's built-in lists for numerical operations.

Important Features

- **Multidimensional Arrays:**
  Efficientstorageandmanipulation of large datasets (e.g., 1D, 2D, 3D arrays).
- **Mathematical Functions:**
  Awide rangeof mathematical operations — linear algebra, Fourier transforms, random number generation, etc.
- **Vectorization:**
  Operationsonentire arrays without explicit loops, making code cleaner and faster.
- **Interoperability:**
  Workswellwithother libraries like Pandas, Matplotlib, TensorFlow, PyTorch etc.

Usage

```
import numpy as np

# Create & Output an array
arr = np.array([1, 2, 3, 4, 5])
print(arr)
```

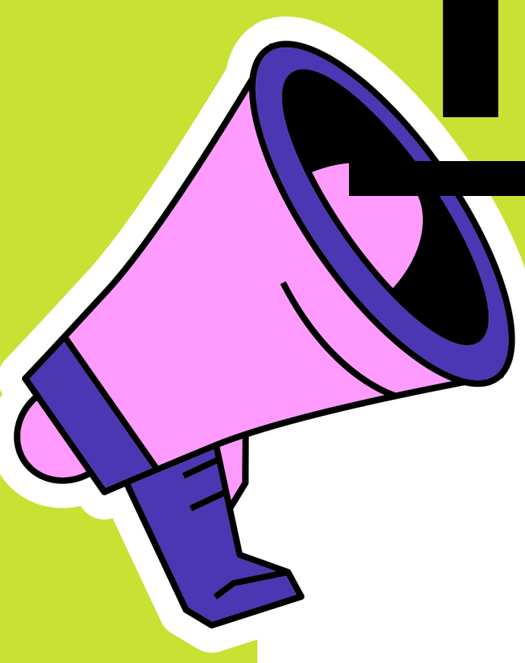
NumPy Arrays vs Python Lists

1. NumPy arrays are faster than Python lists (implemented in C, not pure Python).
2. They store elements in **contiguous memory** (better cache performance).
3. They are also homogenous i.e. all elements have same type.
4. They use **vectorized operations** (no slow Python loops).

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